

Craigslist Clone UML Diagram (For Sale Ads Only)

Simplified UML diagram focused on posting and managing ads in the **For Sale** category.

1. Use Case Diagram

```
%% UML Use Case Diagram for For Sale ads
flowchart TD
    User([User])
    Admin([Admin])
    System((Craigslist Clone))

    User -->|create| UC1[Post For Sale Ad]
    User -->|update| UC2[Edit Ad]
    User -->|delete| UC3[Delete Ad]
    User -->|view| UC4[View For Sale Ads]
    Admin -->|manage| UC5[Moderate Ads]

    UC1 --> System
    UC2 --> System
    UC3 --> System
    UC4 --> System
    UC5 --> System
```

2. Class Diagram

```
classDiagram
    class User {
        +Long id
        +String name
        +String email
        +String passwordHash
        +Date createdAt
        +List<Ad> ads
    }

    class Ad {
        +Long id
        +String title
        +String description
    }
```

```

+Double price
+String condition
+String location
+List<Image> images
+Category category
+Date createdAt
+Date updatedAt
+AdStatus status
}

class Category {
+Long id
+String name
+String slug
}

class Image {
+Long id
+String url
+String thumbnailUrl
}

class AdStatus {
<<enumeration>>
ACTIVE
PENDING
SOLD
REMOVED
}

User --> Ad : owns >
Ad --> Category : belongs to >
Ad --> Image : has many >

```

3. Sequence Diagram – Posting a For Sale Ad

```

sequenceDiagram
    actor User
    participant AngularApp as Angular Frontend
    participant API as Spring Boot API
    participant DB as PostgreSQL
    participant S3 as S3 Storage

    User ->> AngularApp: Fill Ad Form + Upload Images

```

```
AngularApp --> API: POST /api/v1/listings (JSON metadata)
API --> DB: Save Ad (status=PENDING)
API --> S3: Upload Images (presigned URLs)
API --> DB: Update Ad (status=ACTIVE, add image URLs)
API --> AngularApp: 201 Created (Ad details)
AngularApp --> User: Display success confirmation
```

4. Key Entities (For Sale Context)

- **User**: can create, edit, delete ads
 - **Ad**: represents an item for sale
 - **Category**: in this case fixed as "For Sale"
 - **Image**: linked to each ad (stored in S3)
 - **Status**: indicates the lifecycle of an ad (pending, active, sold, removed)
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5. Next Steps

- Extend UML to include **search, filters, messaging, and reporting**.
 - Add **Admin moderation** flow and **approval process** for ads.
 - Include **notifications** for ad approval or expiry.
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(Would you like me to generate the next UML diagram for search and filtering next?)